# Why Inference Latency Explodes Near Full Utilization

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We model a single GPU inference server as an M/M/1 queue. Requests arrive as a Poisson process and service times are exponential.

rho = lambda / mu
W = 1 / (mu - lambda)
Wq = lambda / (mu (mu - lambda))

The server has a service rate of 12 requests per second and sees an arrival rate of 9 requests per second at peak, giving utilization of 0.75.

Limitation: real inference servers batch requests, so this single-queue approximation only illustrates the shape of the latency curve.

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